

DOUGLAS, GA, USA

WMO: 722062

Lat: 31.477N

Lon: 82.861W

Elev: 78

StdP: 100.39

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63842

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.3	0.6	-12.7	1.3	5.2	-10.1	1.6	6.9	8.9	17.5	7.6	12.8	1.5	340	0.349

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	36.2	24.1	35.0	24.1	33.8	23.8	27.5	30.9	26.4	31.0	25.7	30.5	2.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.9	29.7	25.2	20.6	28.0	24.2	19.3	27.3	87.8	31.5	82.5	31.2	79.4	31.1	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.5	6.4	5.4	DB	-3.8	37.2	2.1	1.2	-5.3	38.0	-6.5	38.7	-7.6	39.4	-9.2	40.2	(4)
			WB	-5.6	27.4	1.8	1.0	-6.9	28.1	-8.0	28.7	-9.0	29.3	-10.3	30.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	20.4	11.1	13.1	16.0	19.9	24.0	27.3	28.3	28.2	26.1	21.2	15.5	13.1	(6)
		DBStd	7.16	5.30	4.82	4.61	3.44	2.74	1.85	1.50	1.51	2.14	3.93	4.25	4.98	(7)
		HDD10.0	112	52	22	8	0	0	0	0	0	0	6	25		(8)
		HDD18.3	808	229	156	99	23	2	0	0	0	21	104	174		(9)
		CDD10.0	3893	86	108	194	298	435	519	568	482	347	172	121		(10)
		CDD18.3	1546	4	9	28	71	178	269	309	305	232	108	19	12	(11)
		CDH23.3	14949	22	59	247	713	1667	2752	3311	3066	2084	840	138	51	(12)
(13)	Wind	CDH26.7	6513	0	4	48	206	694	1344	1624	1433	891	252	15	3	(13)
		WSAvg	2.0	2.3	2.3	2.4	2.3	2.1	1.9	1.6	1.6	2.0	2.0	2.1	2.1	(14)
		PrecAvg	1277	122	95	125	60	61	154	150	144	121	91	62	75	(15)
		PrecMax	1638	405	190	273	133	150	268	218	213	411	262	160	150	(16)
		PrecMin	863	20	22	4	3	9	46	51	75	20	13	7	21	(17)
		PrecStd	224	85	44	74	36	35	58	35	36	88	73	43	32	(18)
		DB	25.9	27.3	30.0	32.3	34.7	37.5	37.2	37.4	35.4	32.4	28.1	27.2		(19)
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	19.2	18.8	18.3	19.5	21.7	23.8	24.7	25.0	23.4	22.4	20.2	20.3		(20)
		DB	23.6	25.2	27.8	30.9	32.9	36.0	36.0	36.1	33.7	31.0	27.1	24.9		(21)
		MCWB	18.0	17.8	18.4	19.2	20.7	23.4	24.6	24.7	22.8	21.5	19.0	19.0		(22)
		DB	22.0	22.9	26.3	28.9	32.2	34.1	34.8	34.1	32.6	29.1	25.1	22.7		(23)
		MCWB	17.2	16.6	17.7	18.6	20.5	23.1	24.7	24.6	23.0	20.7	18.0	18.9		(24)
		DB	19.1	21.3	24.0	27.5	30.9	32.8	33.3	32.8	31.5	27.7	22.9	21.3		(25)
		MCWB	16.2	16.1	16.5	17.8	20.7	23.6	24.6	24.6	22.9	20.0	17.4	18.3		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	20.8	20.9	21.2	22.5	24.1	29.0	29.6	27.5	26.1	25.2	23.4	22.3		(27)
		MCDB	22.5	23.5	25.7	26.8	29.3	32.4	33.3	32.1	31.1	28.5	26.7	24.6		(28)
		WB	19.4	19.6	20.4	21.6	23.5	27.3	28.0	26.7	25.4	24.2	22.0	21.0		(29)
		MCDB	22.4	23.1	24.8	26.6	28.4	29.8	31.2	31.7	30.2	27.4	24.2	22.7		(30)
		WB	18.3	18.4	19.5	20.8	22.9	25.7	26.6	26.1	24.8	23.4	20.5	19.9		(31)
		MCDB	20.5	22.3	23.6	25.3	27.7	30.0	31.1	31.1	29.4	26.4	23.1	22.2		(32)
		WB	17.0	17.2	18.5	20.0	22.3	24.9	25.7	25.6	24.3	22.5	18.8	18.5		(33)
(34)	Mean Daily Temperature Range	MCDB	18.9	19.6	22.2	24.6	27.1	30.0	30.7	30.5	28.6	25.7	22.3	21.0		(34)
		MDBR	11.2	11.7	12.4	12.8	11.9	10.7	10.5	9.9	10.2	11.8	12.4	11.0		(35)
		5% DB	12.3	13.2	14.0	14.6	13.5	12.6	11.8	11.7	11.8	12.6	13.6	11.8		(36)
		MCWBR	7.7	7.3	6.0	5.0	3.8	3.6	3.3	3.2	3.5	4.9	6.7	7.3		(37)
		5% WB	10.3	11.1	11.3	11.7	10.1	10.4	10.2	9.8	9.3	8.7	9.5	9.9		(38)
		MCWBR	7.8	7.5	6.1	5.7	3.7	4.5	4.3	3.4	3.3	4.4	6.5	7.5		(39)
		taub	0.325	0.332	0.366	0.387	0.428	0.476	0.510	0.503	0.425	0.377	0.347	0.329		(40)
(41)	Clear-Sky Solar Irradiance	taud	2.512	2.493	2.417	2.350	2.273	2.205	2.110	2.131	2.341	2.445	2.478	2.526		(41)
		Ebn at Noon	902	929	916	904	865	820	790	791	845	867	868	878		(42)
		Edh at Noon	87	97	113	125	136	145	159	154	120	101	90	82		(43)
		RadAvg	3.02	3.62	4.76	5.84	6.42	6.16	6.12	5.56	4.88	4.23	3.35	2.63		(44)
(45)	All-Sky Solar Radiation	RadStd	0.18	0.40	0.43	0.44	0.38	0.41	0.36	0.35	0.40	0.40	0.24	0.24		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	+0.51	+0.75	N/A	N/A	N/A	N/A	N/A	-79	+170	+120

Nomenclature: See separate page